

Bayesian Spatio-temporal analysis of greenhouse gas emissions in Africa

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Abstract

Global warming is a key driver of climate change and greenhouse gases are the main contributors to global warming. This study focuses on greenhouse gas emissions in Africa for 27 years, 1990-2016. Bayesian Spatio-temporal analysis is employed in carrying out this study using R-INLA package. The study aims at understanding the behavior of these emissions from five selected sectors in Africa, and to determine whether a spatial and temporal correlation exists in the emissions. These sectors are agriculture, waste, land-use change and forestry, industry, and energy. Among the eight models formulated, the model that consider space-time interaction appears to perform better than models without the interactions. Results reveal the presence of spatial correlation, temporal correlation, and an increase in these emissions over time.

Declaration

I, the undersigned, hereby declare that the work contained in this research project is my original work, and that any work done by others or by myself previously has been acknowledged and referenced accordingly.



Rebecca Ayodeji Akeresola, May 2020

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List of Abbreviations

GHGs	Greenhouse gases.
CO_2	Carbon dioxide.
CH_4	Methane.
N_2O	Nitrous oxide.
INLA	Integrated Nested Laplace Approximation.
IPCC	Intergovernmental Panel on Climate Change.
UNIDO	United Nations Industrial Development Organization.
CAR	Conditional Autoregressive Model.
DRC	Democratic Republic of Congo.
DIC	Deviance Information Criterion.
$MtCO_2$	Million metric tons of carbon dioxide.
$MtCO_{2e}$	Million metric tons of carbon dioxide equivalent.

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1. Introduction

1.1 Background

The earth is the most suitable planet that can sustain life. Among the various features that make the earth habitable, the atmosphere stands out to be one of the most important and significant features. The earth is held in place by its gravity and surrounded by a layer of gases-air called atmosphere (Babatunde, 2020). The atmosphere which can be referred to as abstract because it cannot be touched, neither can it be seen, is classified into four layers which are the troposphere, stratosphere, mesosphere, and thermosphere. This classification is achieved using the temperature profile.

Each of these layers plays a unique role to make the earth habitable. The sun is the primary source of energy for the earth that warms the earth through radiation. During this process, ultraviolet rays are released from the sun and must not reach the earth's surface because they are harmful to human skin. The stratosphere, which is the second layer of the atmosphere contains ozone molecules that help in preventing the harmful rays from reaching the earth's surface by absorbing them.

The first layer of the atmosphere being the troposphere is where life exists and contains various gases worth 5000 Millions of tonnes (Mt) in which Nitrogen accounts for a total of 78%, oxygen (O_2) 21%, carbon dioxide (CO_2) 0.33%, argon 0.93%, and many other gases (Balasubramanian, 2011). These gases can be classified into greenhouse gases (GHGs) and non-greenhouse gases and are of great importance to human life. For instance, O_2 gas, a non-GHG, is needed for survival as humans and mammals breathe it in during respiration. The importance of this gas cannot be overlooked as its absence marks the end of the human race and life generally on earth.

The GHGs are CO_2 , methane (CH_4), nitrous oxide (N_2O), water vapor, and fluorinated gases. Perfluorocarbons (PFCs), hydro-fluorocarbons (HFCs), and sulphur hexafluoride (SF_6) are examples of fluorinated gases and are human-made. According to the United Nations Framework Convention on Climate Change (UNFCCC), these GHGs excluding water vapor are described as the main GHG emissions called carbon dioxide equivalents (CO_2e). Without these GHGs present in the atmosphere, the average global temperature of the earth would be $-18^\circ C$ (Babatunde, 2020).

Life on earth with such temperature would be uncondusive as most humans, animals, and life activities will not be able to thrive and function well for long under such a situation. The presence of GHGs in the atmosphere helps to raise the average global temperature to $15^\circ C$ as a result of their ability to trap heat that is released by the earth's surface when heated by the sun. This process is referred to as the greenhouse effect hence the name greenhouse gases.

From the point stated above, the GHGs in the atmosphere contributes to making the earth habitable. Just as the overdose of a drug poses more harm than good, the over-concentration of the GHGs in the atmosphere poses more harm than good to the earth, climate, and human. The GHGs occur naturally in the atmosphere however, their concentration in the atmosphere can be altered by human activities grouped into sectors which are agriculture, energy, land-use change and forestry, waste, and industry.

Agricultural activities such as the addition of manure and fertilizer to the soil, cultivation, and the addition of lime to the soil, result in the emission of CO_2 , CH_4 , and N_2O into the atmosphere (Tubiello, 2019). Activities from the energy sector such as the manufacturing of cement, production, transportation, and burning of fossil fuels, coals, natural gas, and oil, contribute to the emission of GHGs (Bruckner et al., 2014). Burning of solid waste, decaying of organic waste and treatment of wastewater are examples of activities from the waste sector that continuously release CO_2 , CH_4 , and N_2O into the

atmosphere (Programme, 2010). Deforestation, which is the felling of trees and burning, conversion of land from its natural use into other purposes, are activities in land-use change and forestry that release GHGs. Activities that are industries such as the production of aluminum emits PFCs as a by-product into the atmosphere, usage of HFCs for refrigeration and air-conditioning activities release HFCs into the atmosphere, and emissions of SF_6 when used as tracers, transformers and electrical equipment (Khalil, 1999).

Research has shown that the concentration of GHGs in the atmosphere has increased over the years as a result of various human activities. For instance every year, over 35 billion tons of CO_2 is released into the atmosphere with 3 to 5 billion tons as a result of tropical deforestation and other land-use change and over 30 billion tons as a result of fossil fuel burning (Council, 2010). Few of the countries which contribute to a large percentage of the total CO_2 emission from cement production and fuel fossils in Africa are South Africa, Egypt, Algeria, Nigeria, Libya and Morocco with 38% from South Africa and a total of 46% from the other five countries (Boden et al., 2011).

Africa accounts for about 4% of the world emission of GHGs making it the continent that has the minimum contribution of GHGs to the world's GHGs emissions (Amadou, 2016). However, Palmer et al. (2019) discovered that tropical Africa had a huge unexpected net carbon emission in 2015 and 2016 respectively. The emissions for 2016 showed that tropical Africa had a more substantial emission than the United States of America, a country among the top emitters of GHGs. This was as a result of land-use change over tropical Africa.

Stabilizing the global average temperature which is the beauty of the presence of GHGs in the atmosphere is constantly changing into being ugly over the years as a result of human activities that increase GHGs concentration in the atmosphere. This increase automatically leads to an increase in the amount of heat energy that is trapped in the atmosphere. The issue of global warming which is an increase in the average temperature of the earth has been attributed to these GHGs because they trap heat energy in the atmosphere. Lozano et al. (2009) did Spatio-temporal causal modeling for climate change attribution and found out that there is a significant impact of GHGs on changes in temperature. The consequences of global warming such as extremely high temperature, drought, heatwaves, melting of ice, rise in sea level, flooding, and many more, are felt by the whole globe, and Africa as a continent is not left out.

1.2 Statement of Problem

Agriculture is key to the sustenance of life. The soil needs to be cultivated, animals need to be reared in order to meet the basic needs of humans. Other sectors such as industrial, energy, health, manufacturing, and so on depend on the agricultural sector for smooth operation and are also crucial to life. The continuous rise in GHGs emissions across the globe has been noticeable from research. Though it is known that Africa contributes a little percentage to world GHGs emissions, Africa has been at the receiving end of the consequence of global warming (IPCC, 2014).

Activities from the land-use sector contribute to the emissions of these gases. During recent decades, there has been a frequent occurrence of drought in tropical Africa as a result of land-use changes (Paeth et al., 2009). Drought is an unfavorable situation for human life because it directly affects agricultural activities negatively. The soil moisture is drastically reduced, the soil texture is affected, and there is a shortage of water, which creates an unpalatable environment for the plants to grow.

According to the Intergovernmental Panel on Climate Change (IPCC), the agriculture sector, especially in West Africa is threatened as the output of crop yield and production has a high tendency of reducing. This is as a result of the large increase in the number of hot days it will experience.

This signals that there will be reduced precipitation and if prolonged, causes drought. The high tendency to reduce crop yield will not leave a good imprint as the demand for food will greatly surpass the availability of food. This paints a perfect picture of food scarcity and if not handled carefully might explode into famine. Other negative impacts of these emissions are flooding, extremely hot temperature, heatwaves, and many more.

Sectors which are energy, land-use change and forestry, waste, industrial, and agriculture all contribute to the increase in the concentration of these GHGs in the atmosphere but the agriculture sector is more at a greater risk of suffering from the consequences of these emissions. As a result of this, there is a need to adopt an appropriate statistical model to analyze the changes in the emissions of these GHGs over space and time in Africa. This will enlighten our understanding of their changes across space, time, and interaction of both in providing useful information that will help to curb the continuous increase of these emissions across the sectors in Africa.

1.3 Aim and Objectives

This essay aims to assess the patterns in GHGs emissions over space and time among the African countries and also by the selected sectors.

The objectives are:

- carry out Bayesian Spatio-temporal analysis on the GHGs emissions
- develop various models and select the best of the models
- determine the spatial and temporal correlation in the GHGs emissions across the African countries and their sectors

1.4 Scope and Limitation of The Study

Among the seven continents around the globe, this study is limited to the African continent. The selection of these countries is based on the strict requirement of the Bayesian spatial modeling technique employed where a country must share boundaries with another country. As a result of this, the study is limited to 44 African countries whose data are available within the study period.

2. Literature Review

This chapter focuses on the previous researches that are related to this study and helpful for a better understanding of the study. A review of the literature on GHGs emissions and application of Bayesian Spatio-temporal analysis and modeling are discussed in this chapter.

2.1 Greenhouse Gases and Their Emissions

By studying the impact of GHGs emissions on crop production in West Africa between 2000-2016, Osabohien et al. (2019) showed that there was a significant decrease in the yield of crops. Their study sheds more light on how negatively the agricultural sector is being affected by the emissions of GHGs into the atmosphere especially in West Africa.

Time series analysis was employed by Doğan and Karakaş (2019) in a study on the nexus of GHGs emissions (CO_2) and the agriculture sector. In the study, data on GHGs emissions from this sector was considered from the year 1991 to 2017, with the case study focusing on China and Turkey. The variables considered were poultry, cattle, and small ruminant husbandry, and paddy production. Their results revealed that all variables considered had a significant effect on CO_2 emissions in China, while for Turkey, all variables excluding poultry husbandry had a significant effect on CO_2 emissions.

Charkovska et al. (2019) explored the spatial distribution of the GHGs emissions, CO_2 , CH_4 , and N_2O , from Poland's agriculture sector. The spatial component is the municipalities in Poland. For the study, the data on animal stock for the year 2010 were obtained. The study results showed that the model that considered the spatial component outperformed the model that neglected the spatial component. As a result, the model with the spatial component produced better estimates of the GHGs emissions from the agriculture sector. The study revealed the variations and patterns in the GHGs emissions across the study area.

A study by Li et al. (2020) on the impact of energy prices on CO_2 emissions in China used spatial econometric method relying on data from the year 2002 to 2006. The study discovered that the energy price had a negative impact on the emissions of CO_2 . This study did not consider the interaction between space and time but focused only on the spatial component as it employed the cumulative emissions from 2002 to 2006.

Another research focused on the relationship between GHGs emissions, energy consumption, and economic growth by Yusuf et al. (2020). The study covered the six oil-producing countries of Africa, while the GHGs covered are CO_2 , CH_4 , and N_2O from 1970-2016. Panel autoregressive distributed lag model was employed for the analysis, and results revealed that in the long run, the effect of the economic growth on CH_4 and CO_2 were significantly positive, and only for CH_4 in the short run. Interestingly, on the other hand, the effects of energy consumption on the three GHGs were insignificantly positive both in the long and short run.

It is known that the waste sector contributes to the global emissions of GHGs. A study by Ngwabie et al. (2019) serves as a confirmation of the stated fact. Considering two different geographical locations, which are the south-west region and north-west region within the country of Cameroun, they focused on the open dumpsite that received solid waste daily. The dumpsites were monitored to quantify the emissions of GHGs for a period of 16 months, May 2015 to August 2016. At the end of the time frame, it was recorded that the CO_2 , CH_4 , and N_2O were emitted from the dumpsites, but the amount of emissions varied between the locations.

There are numerous open dumpsites with waste across African countries which cause more harm because the amounts of GHGs being released into the atmosphere, cause the concentration of these GHGs in the atmosphere to increase. It is therefore advisable to eliminate open dumpsites of waste. Other literature that focuses on activities from the waste sector in relation to GHGs emissions is [dos Muchangos et al. \(2019\)](#) of which the case study was Maputo, the capital of Mozambique, [Maqhuzu et al. \(2019\)](#) with a case study of Zimbabwe, [Mohsen et al. \(2019\)](#) with a case study of Ontario, Canada, and [Maria et al. \(2019\)](#) with a case study of Angola.

From the selected few works of literature related to GHGs emissions, some with case study different from Africa considered analysis of GHGs emissions employing methods such as time series and spatial analysis. Others with a case study on the African continent looked at GHGs from various sectors. Sectors such as agriculture, waste, energy, and land-use with a selection of a few African countries. Few or no literature is found on the analysis of GHGs emissions in Africa using Bayesian Spatio-temporal modeling. Also, most literature considers CO_2 , CH_4 , and N_2O with little or no concentration on the fluorinated gases, which are also GHGs. This study is to carry out a Bayesian Spatio-temporal analysis of GHGs emissions in Africa, with fluorinated gases inclusive and therefore bridge the gap in previous works of literature.

2.2 Bayesian Spatio-temporal Analysis

The Bayesian Spatio-temporal analysis is an aspect of analysis in statistics that serves as an extension of spatial analysis and has been used in various fields of research most notably in the health sector for disease mapping.

[Wang et al. \(2008\)](#) modeled the prevalence of *Schistosoma japonicum*, an infectious disease using a Bayesian Spatio-temporal approach. The data used for the study contained the report of the infectious disease from 1995 to 2004 of 114 villages in the southeastern part of China with socio-economic variables and environmental variables serving as covariates. Seven models were developed in which three models considered spatial effect and temporal effect, three models considered the space-time interaction in which the spatial correlations evolved and the last model did not take into account the spatial effect. Their result revealed that the models with space-time interaction outperformed other models, but the best model was the one that considered the environmental covariates with space-time interaction. If they had settled with models that did not consider space-time interactions, they would have loosed vital information that can only be accessed when space-time interactions are considered. This shows the importance of Spatio-temporal modeling.

The Bayesian Spatio-temporal analysis was adopted by [Raghavan et al. \(2016\)](#) to analyze Rocky Mountain spotted fever (RMSF), a disease that mostly affects adults and children. The case study was four states in the United States, and the counties in the states served as the spatial component. The data used for the study were the reported cases of RMSF at the county level from the year 2005-2014 for the four states. The model considered the space-time interactions and part of their results showed that over time, the RMSF surfaced in new geographic locations at a steady pace.

Using the Bayesian approach, [Briz-Redón and Serrano-Aroca \(2020\)](#) carried out a study on the Spatio-temporal analysis to explore the effect of temperature on COVID-19 early evolution in Spain. The study utilized the COVID-19 and temperature data for the provinces in Spain within a time frame of two weeks. The model considered the spatially structured effect, spatially unstructured effect, temporal structured, and unstructured effect with which the computing was done using Integrated Nested Laplace Approximation (INLA). The result revealed that in the provinces with warmer temperature (mean,

maximum, and minimum), there was no evidence of reduced cases of COVID-19. Though the study took into consideration the spatial and temporal component, they did not consider the effect of the interaction of space and time in their model. Besides, only one model was developed. Developing more than one model provides the opportunity for comparison to select the best model.

Spatio-temporal analysis has been employed in climate research. A study by [Mabaso et al. \(2006\)](#) considered the role of climate in inter-annual variation of malaria incidence by employing Spatio-temporal analysis, with Zimbabwe as a case study. Climatic variables and malaria incidence data for the years 1988 to 1999 were used with the districts of Zimbabwe serving as the spatial unit. The Bayesian approach was employed, and three models were developed. The first model did not consider the spatial effect, the second model considered the spatial effect and temporal effect while the third model considered the space-time interaction. Bayesian computing was done using Markov chain Monte Carlo (MCMC) and interestingly, the third model outperformed the other two. Results showed that climate is a driving force for the year to year variation in malaria incidence.

Similar work is done by [Chirombo et al. \(2020\)](#), where the impact of climate and non-climate factors on childhood malaria incidence was studied using Spatio-temporal modeling. The case study was Malawi and the data used was from 2004 to 2017. The model developed considered the space-time interaction, computation, and the results revealed that there was a significant impact of climatic and environmental factors on malaria incidence. Also, between the climatic zones, a minimal increase in malaria incidence was noticed during the study period and was different across the country.

[Hundera et al. \(2020\)](#) did a study on Spatio-temporal analysis of land-use and land-cover dynamics and its implication to GHGs. The case study considered was a district in Ethiopia called Adama. The data used was GHGs emissions for the years 1986, 2000, and 2014. The results revealed that as a result of human activities, lands were converted from their natural use for human purposes. This study did not consider space-time interaction but rather obtained the individual spatial distribution of land-use and land cover for each year.

A research work by [de Rivera et al. \(2020\)](#), explored the causality of forest fires in Europe using Spatio-temporal model. The data used for the study contained information on forest fires, socio-economic variables, and environmental variables from 2012 to 2016. The model built considered fixed effect and random effects. The random effects are space and time, which accounted for the spatially structured, spatially unstructured, temporally structured, temporally unstructured, and the interaction between space-time. The temporally unstructured and spatially unstructured effect was the choice they selected for the interaction. Bayesian computation was carried out using INLA and results showed that the temporal effect, spatial effect, interaction of space-time, socio-economic variables, and environmental factors affect the occurrence of forest fires. Applying the Bayesian approach gave them a better understanding of the causes of forest fire than when a traditional approach is employed.

3. Data and Methodology

3.1 Data Description

The data used for the study were obtained from the World Resource Institute (WRI), <https://www.wri.org>, responsible for the collection and keeping of climatic data. The yearly data were on the GHGs emissions in Africa and by its sectors from 1990-2016, 27 years. Due to the strict requirements of the Spatio-temporal analysis, an entity in the spatial component must share boundaries with one or more entities. The data on African countries sharing boundaries with one or more African countries were extracted. A total of 44 African countries were considered for this study. The GHGs accounted for in the data are CO_2 , CH_4 , N_2O , and fluorinated gases. These gases are reported in their Million metric tons of CO_2 equivalent ($MtCO_2e$), and the sum of their emissions serves as the total GHGs emissions in each country. In this study, the sectors considered are agriculture, waste, land-use change and forestry, industry, and energy. The countries represent the spatial component, while the years serve as the temporal component. Figure 3.1 presents a ratio map of Africa, showing the countries included in the study.



Figure 3.1: Map of Africa showing the 44 countries considered for the study. The country in white (South-Sudan) was excluded from the study due to lack of data

3.2 Spatial Analysis

Spatial means space or a geographical location and can be of various dimensions such as Continent, Country, Region, State, District, Counties, Street, Local government, and any other entity with the same attributes as the examples listed. Given a parameter of interest whose basis of collection and

representation is the geographical locations listed above, or whose geographical location in space is known, such a parameter of interest is termed a spatial data.

There are so many parameters of interest that are represented by the geographical location, examples such as HIV prevalence in Nigeria with the states of the country serving as the spatial component, malaria prevalence in Rwanda, with the districts in Rwanda serving as the spatial component, the number of people with lung cancer in Ohio with the counties serving as the spatial component, the outbreak of COVID-19 virus in the world, with either the continents or countries serving as the spatial component and so on. Data from these forms can be classified into three types geostatistical, point pattern, and areal data (Blangiardo and Cameletti, 2015).

Data with locations represented with coordinates (latitude, longitude) and sometimes altitude inclusive are known as geostatistical data. Areal data, on the other hand, are data with locations that are of dimensions such as country, states, district, counties, and regions. The data used for this study are areal because the dimension of the location that serves as the spatial component is the countries in Africa. Locations represented with points where the parameter of interest in the data occurs are termed as point pattern data. These spatial data, when analyzed using various techniques, are known as spatial analysis. The behaviors and patterns of these data and how they change can be studied across space.

3.3 Temporal Analysis

Temporal means time. Any data that is collected with respect to time is referred to as temporal data. Temporal can be called time-series and temporal data, time-series data. The time components can be of various dimensions, such as years, months, quarterly, weeks, days, hours, minutes, and seconds. Temporal analysis reveals the behavior of the observations of temporal data over time. Patterns in the temporal data can easily be identified, trends that can either be positive or negative, increasing or decreasing can be seen.

3.4 Spatio-temporal Analysis

It is essential to know the importance of the need for Spatio-temporal analysis having described what spatial and temporal analysis means. The latter explains the data behavior over time but does not account for the changes in these behaviors over space. The former, on the other hand, explains the data behavior across space but does not give an account of these changes over the time component. These two concepts have been used repeatedly in literature and research but fail in situations where vital information can be obtained from the data when space and time interaction is taken into consideration. The marriage between the spatial analysis and temporal analysis gives birth to Spatio-temporal analysis and as well takes into account the interaction of space and time components.

3.4.1 How The Spatio-temporal Analysis Works. Focusing on the spatial component, there is a possibility that a particular unit i in the spatial component may share boundaries with one or more other units within the spatial component. These units are referred to as the neighbors of unit i . The spatial component of the data used in this study comprises the 44 African countries selected that satisfy the boundary rules. The temporal component used in this study is 27 years from 1990-2016. Unlike the spatial component, there is no rule binding the temporal component in the Spatio-temporal analysis.

The spatial and temporal components are denoted as:

$$s = s_1, s_2, s_3, \dots, s_{44}$$

$$t = t_1, t_2, t_3, \dots, t_{27}$$

The information on the structure of the neighborhood is crucial because it is used to form the adjacency matrix that is used in the model to account for the spatial structure. It is a symmetry square matrix with zeros and ones, and with the dimensions equal to the number of entries in the spatial component. The adjacency matrix for our study is of dimension 44×44 since the total number of countries considered is 44.

If s_2 and s_3 are neighbors of s_1 , then,

$$s_1 s_2 = s_1 s_3 = 1$$

$$\implies s_2 s_1 = s_3 s_1 = 1.$$

If the reverse is the case i.e. s_1 do not share boundaries with s_2 and s_3 , then,

$$s_1 s_2 = s_1 s_3 = 0$$

$$\implies s_2 s_1 = s_3 s_1 = 0.$$

It is impossible for a country to share boundary with itself. Therefore:

$$s_1 s_1, s_2 s_2, s_3 s_3, s_4 s_4, \dots, s_{44} s_{44} = 0$$

This process is the manual way of forming the adjacency matrix and can be time-consuming when we have large data such as in our case.

An alternative process of obtaining the adjacency matrix which was employed for this study is by first obtaining a shapefile from the spatial component by using packages in R and then converting it into an adjacency matrix using the INLA package in R software. INLA is used for this analysis because it is used for Bayesian computing and produces results faster than when MCMC is used. Relevant literature like [Schrödle and Held \(2011\)](#); [Alegana et al. \(2016\)](#); [Briz-Redón and Serrano-Aroca \(2020\)](#), and many more that are not listed as well, have employed INLA for Bayesian computing. The successful creation of the adjacency matrix paves the way for formulating different models in Spatio-temporal analysis. The approach employed in the various models' formulations is the Bayesian approach.

3.5 Why Bayesian Approach?

In statistics, there are two schools of thought, which are the frequentist approach, also known as traditional statistics, and the Bayesian approach. The definition of probability in the frequentist approach is based on the repeatability ([Blangiardo and Cameletti, 2015](#)). Unlike tossing a coin and rolling a fair die, which can be repeatable, most of all real-life issues are not repeatable, thus making the frequentist approach not suitable for such events. The Bayesian approach performs well where the frequentist approach fails. It helps to account for the uncertainties in the data, parameters, and models ([Wikle and Hooten, 2006](#); [Vanem et al., 2014](#)). It is also applied to observed data where the parameter is an

unknown quantity and it is needed to determine their prior distribution in order to obtain the posterior distribution given as:

$$\mathcal{P}(\theta|y) \propto \mathcal{P}(y|\theta) \times \mathcal{P}(\theta).$$

$\mathcal{P}(\theta)$ is the prior distribution that tells us what is known beforehand about the parameter θ and $\mathcal{P}(y|\theta)$ is the likelihood function. When the data (y) has been observed, the posterior distribution gives information about the estimate $\hat{\theta}$ of the parameter θ , through the posterior mean, standard deviation, mode, and quantiles. The 95% credible interval (CI) is given by the 0.025 and 0.975 quantiles. The estimates whose 95% CI does not include zero are said to be statistically significant.

3.5.1 Prior Distributions. The Bayesian approach employed provides the avenue to select prior distributions for the components of the model formulation. The components involved in the models are unstructured spatial random effect, structured spatial random effect, unstructured temporal random effect, and structured temporal random effect. The conditional autoregressive model is selected as a prior distribution for the structured spatial random effect. A second-order random walk is selected as a prior for the structured temporal random effect, and an independent and identically distributed normal prior was considered for the unstructured spatial and temporal effects.

3.5.2 Normal Prior. The unstructured spatial and temporal components assume that there is no dependency among the spatial units and time points. They are modeled as independent and identically distributed (iid). Identical because the random variables are taken from the same probability distribution. The unstructured spatial effect u_i is given as $u_i \sim \mathcal{N}(0, \sigma_u^2)$ while the unstructured temporal effect ϕ_j is given as $\phi_j \sim \mathcal{N}(0, \sigma_\phi^2)$. $\mathcal{N}$ indicates that the effects are normally distributed while σ_u^2 and σ_ϕ^2 are the variances of the unstructured spatial and temporal effect respectively.

3.5.3 Conditional Autoregressive Model (CAR). Spatial autocorrelation exists when the boundary effect is taken into consideration in a model. The spatial random effect error of the model has a mean 0 and a variance-covariance matrix Σ i.e. $z \sim \mathcal{N}(0, \Sigma)$. To account for this spatial autocorrelation, a statistical model which is the CAR is employed. In CAR, the variance-covariance matrix is obtained through the inverse of the precision matrix:

$$\begin{aligned}\Sigma^{-1} &= (\sigma^2)^{-1}(\mathcal{I} - \rho W)D \\ \Sigma &= \sigma^2(\mathcal{I} - \rho W)^{-1}D\end{aligned}$$

where Σ^{-1} = precision matrix, $\mathcal{I}$ = identity matrix, ρ = autocorrelation parameter, W = adjacency matrix, D = diagonal matrix, and Σ = variance-covariance matrix.

The spatial random effect error is then given as:

$$z \sim \mathcal{N}(0, \sigma^2(\mathcal{I} - \rho W)^{-1}D)$$

The CAR model is given as:

$$s_i | s_j, i \neq j \sim \mathcal{N}\left(\frac{1}{N(i)} \sum_{j=1}^n s_j, \frac{\sigma^2}{N(i)}\right)$$

where s_j = neighbors of s_i , and $N(i)$ = number of the neighbors of i .

Since the data considered for this study is areal, through CAR, the Besag model is used to identify the spatial structure effect (Besag et al., 1991). The Besag–York–Mollié (BYM) model Besag et al. (1991) is used to specify the model with the combination of spatially structured (CAR) and spatially unstructured effect (iid).

3.5.4 Random Walk. There is a dependency between the current time-step and the previous time-step in a random walk, also referred to as drunkards walk. It helps to account for temporal autocorrelation. That is, the correlation between an observation at a current time and the same observation at a previous time. The first-order random walk is given as

$$y_t = y_{t-1} + \varepsilon_t$$

where y_t is the observation at the current time, y_{t-1} is the observation at the previous time, and ε_t is the step size at the current time.

The second-order random walk is given as

$$\begin{aligned} rw2 : y_t &= y_{t-1} + v_t \\ \text{where } v_t &= v_{t-1} + \varepsilon_t \\ y_t &= y_{t-1} + v_{t-1} + \varepsilon_t \\ \text{but } y_{t-1} &= y_{t-2} + v_{t-1} \implies v_{t-1} = y_{t-1} - y_{t-2} \\ y_t &= y_{t-1} + y_{t-1} - y_{t-2} + \varepsilon_t \\ &= 2y_{t-1} - y_{t-2} + \varepsilon_t \end{aligned}$$

The random walk order 2 is given as $y_t | y_{t-1}, y_{t-2} \sim \mathcal{N}(2y_{t-1} - y_{t-2}, \sigma^2)$

3.6 Integrated Nested Laplace Approximation

The Integrated Nested Laplace Approximation (INLA) is used for Bayesian computation and is applied to latent gaussian models. The goal of the INLA is to obtain the posterior marginals of the model parameters and the hyperparameters using Laplace approximation and numerical integration. The posterior mean, standard deviation, mode, and quantiles are obtained from the posterior marginals. The Latent Gaussian models which are a subset of a structured additive model can be applied to regression models, spatial models, Spatio-temporal models, geostatistical models, states space models, and so on.

In the Latent Gaussian models, the parameters, $\mathbf{x}$ of the model taken to be Gaussian are known as a latent field while the hyperparameters, θ of the models are taken to be non-Gaussian. The hyperparameters, such as precision, variance, dispersion, and correlation, control the effects in the model. These include random effect, fixed effect, linear and non-linear effect, temporal effect, and so on. Just as the Bayesian approach ensures that we obtain information about the posterior distribution through the likelihood and the prior distribution, the posterior density of the Latent Gaussian model that combines the latent fields and the hyperparameters is given as :

$$\pi(\mathbf{x}, \theta | y) \propto \pi(\theta) \pi(\mathbf{x} | \theta) \prod_{i \in \tau} \pi(y_i | x_i, \theta)$$

where $\pi(\mathbf{x}, \theta | y)$ is the posterior density, $\pi(\theta), \pi(\mathbf{x} | \theta)$ are the prior density, and $\prod_{i \in \tau} \pi(y_i | x_i, \theta)$ is the likelihood function.

The posterior marginals of the hyperparameters and the latent fields given as $\pi(\theta|y)$ and $\pi(x_i|y)$, respectively, can be obtained using Laplace approximation. To approximate these marginals, the conditional distributions can be approximated, and for the hyperparameter is given as:

$$\pi(\theta|y) = \frac{\pi(x, \theta, y)}{\pi(x|y, \theta)}$$

Similarly, the latent fields are given as:

$$\pi(x|y) = \frac{\pi(x, \theta, y)}{\pi(x, y|\theta)}$$

A simplified Laplace approximation is employed and Taylor's series expansion for the numerator and denominator is obtained. The posterior marginals are then given as:

$$\begin{aligned}\pi(x_i|y) &= \int \pi(x_i|\theta, y) \pi(\theta|y) d\theta \\ \pi(\theta_j|y) &= \int \pi(\theta|y) d\theta_{-j}\end{aligned}$$

where $\pi(x_i|y)$ is the posterior marginal of latent field, $\pi(\theta_j|y)$ is the posterior marginal of hyperparameter, and $\pi(\theta|y)$ is nested in $\pi(x_i|y)$

3.7 Model Formulation In Spatio-temporal Analysis

The logarithmic transformation of the GHGs emissions is used for the analysis. This is done to make the data follow a normal distribution.

$$y_{ij} = \log(g_{ij})$$

where $i = 1, 2, 3, \dots, 44$, $j = 1, 2, 3, \dots, 27$ and g_{ij} = GHGs emission of country i at year j .

y_{ij} which serves as the dependent variable is represented by a linear predictor η_{ij} , that takes different forms based on the model, and is assumed to follow a normal distribution with parameters μ and σ^2 . i.e. $y_{ij} \sim \mathcal{N}(\mu, \sigma^2)$.

Eight models were formulated for the analysis of the total GHGs emissions and also by the sectors. Model 1, 2, 3, and 4 consider the various possible forms that the temporal effect and spatial effect can take without considering the space-time interactions. Models 5, 6, 7, and 8 are extensions of Models 1, 2, 3, and 4 that take into consideration, the possible various space-time interactions. While the intercept is the fixed effect, the spatial and temporal effects in all the models are random and can be viewed from two different angles, which are structured and unstructured.

Spatially structured means that there is a dependency between countries and their neighbors. In contrast, spatially unstructured means that there is no dependency and each country is independent of other countries. Temporally structured implies that the time points are dependent while temporally unstructured implies that the time points are independent.

3.7.1 Model 1. Spatial (structured) + Temporal (unstructured)

$$\eta_{ij} = b_0 + s_i + \phi_j$$

where

b_0 = intercept (average GHGs emission of the study area)

s_i = spatially structured

ϕ_j = temporally unstructured

3.7.2 Model 2. Spatial (unstructured) + Temporal (structured)

$$\eta_{ij} = b_0 + u_i + \gamma_j$$

where

u_i = spatially unstructured

γ_j = temporally structured

3.7.3 Model 3. Spatial (structured) + Temporal (structured)

$$\eta_{ij} = b_0 + s_i + \gamma_j$$

3.7.4 Model 4. Spatial (structured + unstructured) + Temporal (structured + unstructured)

$$\eta_{ij} = b_0 + u_i + s_i + \phi_j + \gamma_j$$

3.7.5 Model 5. Spatial (unstructured) + Temporal (unstructured) + space-time interaction

It assumes that the space-time interaction to be spatially and temporally unstructured. However, since we know that it is impossible to have this kind of model in reality, we do not select this model; instead, we use it as a reference.

$$\eta_{ij} = b_0 + u_i + \phi_j + \delta_{ij}$$

where

δ_{ij} = spatially and temporally unstructured interaction

3.7.6 Model 6. Spatial (structured) + Temporal (unstructured) + space-time interaction

It assumes that space-time interaction is spatially structured and temporally unstructured. This implies that the spatial structure is independent of the other time points (years).

$$\eta_{ij} = b_0 + s_i + \phi_j + \delta_{ij}$$

where

δ_{ij} = spatially structured and temporally unstructured interaction

3.7.7 Model 7. Spatial (unstructured+structured) + Temporal (unstructured+structured) + space-time interaction

The spatial structure and temporal structure interact in this model. This implies that the spatial structure is not independent of the other time points (years), and the temporal structure is not independent of the other areas (countries).

$$\eta_{ij} = b_0 + u_i + s_i + \phi_j + \gamma_j + \delta_{ij}$$

where

δ_{ij} = spatially structured and temporally structured interaction

3.7.8 Model 8. Spatial (structured) + Temporal (structured) + space-time interaction

It assumes that space-time interaction is spatially structured and temporally structured. This implies that the spatial structure is not independent of the other time points (years) and the temporal structure is not independent of the other areas (countries).

$$\eta_{ij} = b_0 + s_i + \gamma_j + \delta_{ij}$$

where

$$\delta_{ij} = \text{spatially structured and temporally structured interaction}$$

3.8 Model Comparison and Selection

Methods commonly employed in the model selection are Akaike information criterion (AIC), Bayesian information criterion (BIC), Watanabe-Akaike Information Criterion (WAIC), and Deviance Information Criterion (DIC). The DIC is used as the basis of selecting the best model out of all the eight models because it is mostly employed in Spatio-temporal analysis for model selection. It is denoted as:

$$DIC = pD + \bar{D}$$

where

$$pD = \text{effective number of parameters, } \bar{D} = \text{mean deviance}$$

Alternatively,

$$DIC = D(\bar{\theta}) + 2(pD)$$

where $D(\bar{\theta})$ is the deviance at mean and is given as:

$$D(\bar{\theta}) = \bar{D} - pD$$

The model that has the smallest DIC value is termed as the best model that fits the data. The Spatio-temporal analysis as explained is carried out on GHGs emissions data and also on GHGs emissions from the five sectors. The best model for each of the six analysis is reported and interpreted as appropriate.

4. Results and Discussion

This chapter presents all the results obtained from the Spatio-temporal analysis using the Bayesian approach. We employed the INLA package in R software to compute the analysis task. The analysis is classified into two sections: the analysis of GHGs emissions and the sectors, respectively. The analysis of GHGs emissions by sector focuses on five sectors which are agriculture, land-use change and forestry, waste, industry, and energy. The dimension of the data-set is 44×27 , making a total of 1,188 observations, and a time frame of 1990-2016, 27 years for the GHGs emissions was captured for 44 African countries.

Due to the large size of the data-set, the temporal trend of the GHGs emissions and by their sectors for each country across the 27 years was done, in order to give an insight into what the data of the GHGs emissions and by their sector looks like.

4.1 Spatio-temporal Models

Eight models, as described in the methodology, are developed for the total GHGs emissions and the emissions from the five sectors. The model with the least DIC value is considered as the best model that fits the data. Table 4.1 shows the summary of the developed models with their respective DIC values. The space-time interactions are not considered for models 1, 2, 3, and 4. Models 2, 3, and 4 for all the sectors and the total GHGs emissions have similar DIC values implying that they have the same performance. These models outperform model 1 that takes into account the unstructured temporal effect and structured spatial effect. It implies that the models perform better when the temporal effect is taken to be structured or a combination of both structured and unstructured than when only the unstructured effect is taking into consideration.

Models 5, 6, 7, and 8 reveal the importance of capturing the space-time interactions as they all outperform the models that did not capture the space-time interactions except for model 6. It is outperformed by models 2 to 4 in the industrial and energy sector and by models 1 to 4 in the waster sector and total GHGs emissions. The interaction for models 7 and 8 is between the spatial structure and temporal structure with model 7 performing better than model 8 and all other models across the five sectors and the total GHGs emissions. This implies that model 7 is the model that fits the data best because it has the least DIC values.

Table 4.1: Model summary for the GHGs emissions and by the sectors. S= Spatial, T= Temporal, u= unstructured, s= structured, and int= interaction

Models	Model Description	DIC values: Emissions by Sectors & Total GHGs					
		Agric	Industry	Energy	Land-use	Waste	Total GHGs
1	S (s) + T (u)	-1034.76	833.14	1021.39	3046.13	179.89	2185.37
2	S (u) + T (s)	-1064.14	793.54	978.78	3013.67	137.71	2173.21
3	S (s) + T (s)	-1064.11	793.57	978.78	3013.65	137.75	2173.33
4	S (u+s) + T (u+s)	-1062.49	794.42	979.67	3013.84	139.16	2173.71
5	S (u) + T (u)+ int	-1247.81	473.17	662.06	2693.02	-179.18	1842.42
6	S (s) + T (u) + int	-1239.05	830.46	1016.8	3045.67	204.57	2193.42
7	S (u+s) + T (u+s) + int	-2506.04	-1414.5	-3265.55	810.01	-3783.73	148.8
8	S (s) + T (s) + int	-2488.2	-1405.13	-3248.89	846.74	-3775.03	152.09

What the seventh model translates to is that among the 44 African countries, the temporal patterns of total GHGs emissions of each country and from their five sectors are dependent on the temporal patterns of the emissions of the neighboring countries. The estimates of the intercept from model 7 are reported in Table 4.2 and are considered statistically significant since zero is not included in their 95% credible interval (CI). This implies that the average GHGs emissions for the 44 African countries are 38.0614 $MtCO_2e$ while for the agriculture, industry, energy, land-use, and waste sectors are 7.5315, 1.7244, 7.5398, 6.2414, and 1.3172 $MtCO_2e$, respectively.

Table 4.2: Result for the estimates of the intercept for model 7. SD= standard deviation

	Estimates of the intercept of model 7			
	Mean	SD	0.025 quantile	0.975 quantile
GHGs emissions	3.6392	0.0087	3.6225	3.6558
Agriculture	2.0191	0.0066	2.0065	2.0317
Industry	-0.5449	0.0033	-0.5514	-0.5383
Energy	2.0202	0.0063	2.0082	2.0322
Land-use	1.8312	0.0106	1.8105	1.852
Waste	0.2755	0.0061	0.2638	0.2873

The other results reported in this chapter are the Spatio-temporal plot and model-based temporal trend based on model 7 for the five sectors and total GHGs emissions.

4.2 Agricultural Sector

4.2.1 Model-Based Temporal Trend. Figure 4.1 presents the structured temporal pattern for GHGs emissions from the agriculture sector of the 44 African countries. The upper and lower black lines in the temporal plot of all the sectors represent the 95% CI while the blue line represents the posterior mean. The closeness of the CI to the mean estimate indicates a small standard deviation of the estimates. The plot reveals that there were stable emissions of GHGs from the agricultural sector between 1990 and 1996. Since 1996, there has been a continuous increase in the emissions of GHGs up to 2016.

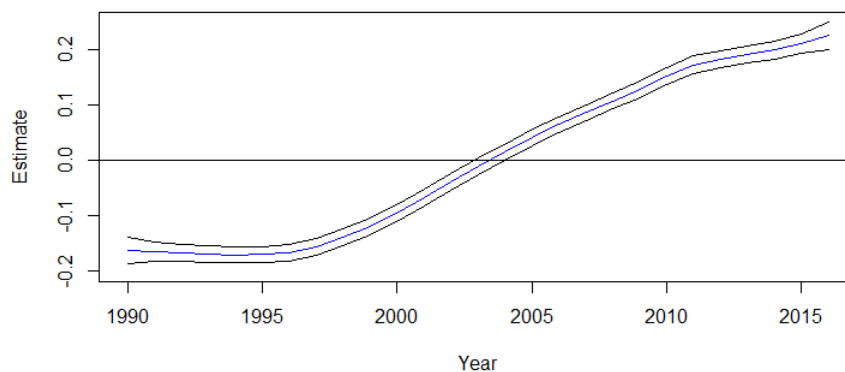


Figure 4.1: Temporal trend of structured effect showing the posterior mean with 95% CI of the GHGs emissions from the agriculture sector.

4.2.2 Spatio-temporal Plot. Figure 4.2 shows the effect of the interaction of space and time on the GHGs emissions from the agriculture sector of each country over the study period. The behavior of the emissions across space and time are revealed. It is observed that from 1990-2016, countries that are estimated to emit a high amount of GHGs from their agriculture sector are Nigeria, Angola, Tanzania, Ethiopia, Sudan, and South-Africa.

Kenya had a high tendency to emit GHGs between 1990-1991 from their agriculture sector. A reduction in this tendency is observed between 1992-2003 but later picked up again in the year 2004 up until 2016. Egypt, DRC, Mali, Uganda, Chad, and Niger were estimated to have a high amount of

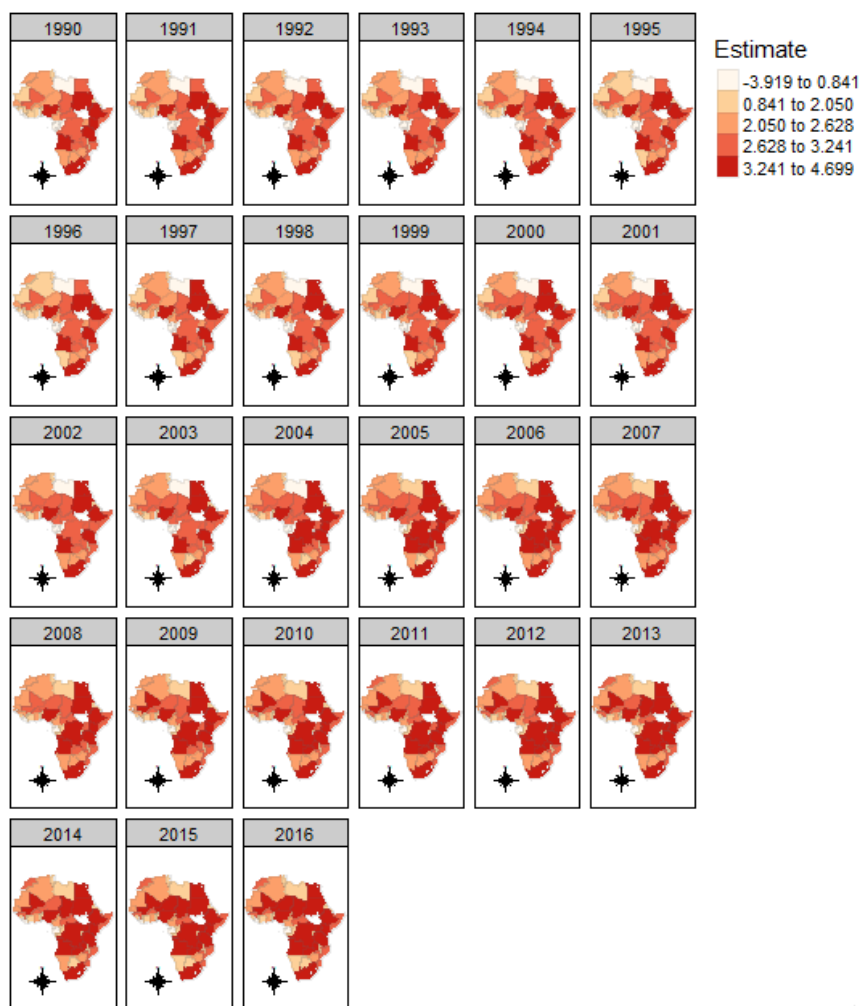


Figure 4.2: Spatio-temporal plot of the posterior mean of the GHGs emissions from the agriculture sector with spatially structured and temporally structured interaction

GHGs emissions in the years 1997, 2004, 2009, 2012, 2013, and 2015, respectively. The emissions from the agriculture sector of these countries maintained the tendency until 2016. Fluctuations are observed in the estimated emissions from the agriculture sector of Zambia but maintained the high estimated emissions of GHGs from 2010 up till 2016.

Countries whose agriculture sector least contributes to GHGs emissions in Africa are Gabon, Equatorial

Guinea, Rwanda, Liberia, Burundi, Djibouti, and Libya. Interestingly, it is noticed that Libya fell out of this category in the year 2005 up until 2016 because of an increase in the estimated GHGs emissions from its agriculture sector.

It is observed that in Northern Africa, the estimated GHGs emission from the agriculture sector in Egypt is more than the estimated emissions of other Northern African countries. This may be due to Egypt being the only Northern Africa country producing rice (Oteng and Sant'Anna, 1999). Rice-production is an agricultural activity known to emit CH_4 and N_2O (Yao et al., 2017; Maraseni et al., 2009).

It is also revealed that in 2016, the agricultural sectors from most of the countries in West, East, Central, and Southern Africa had their GHGs emissions estimates belonging to the top category. Interestingly, these regions constitute sub-Saharan Africa and it is known for intense agricultural practices. Furthermore, it is evident that over the time frame, there has been an increase in GHGs emissions from the agriculture sector of the countries in Africa.

Generally, the effect of the boundary countries shares, on the GHGs emissions from the agriculture sector might not be evident over a short period. However, these shared boundaries' impacts tend to unfold over a long period, as revealed by the Spatio-temporal plot. As a result, the chances of Namibia, Botswana, Zimbabwe, Mozambique, and Malawi, countries within the Southern Africa region, and the remaining countries in West, Central, and East Africa having high emissions of GHGs from the agricultural sector over time are high.

4.3 Industrial Sector

4.3.1 Model-Based Temporal Trend. Figure 4.3 presents the structured temporal pattern for GHGs emissions from the industrial sector of the 44 African countries. The plot reveals that from 1990, the emissions of GHGs from the industrial sector in Africa have been on a continuous rise up till 2016. The closeness of the 95% CI of the estimates indicates how small the standard deviations are.

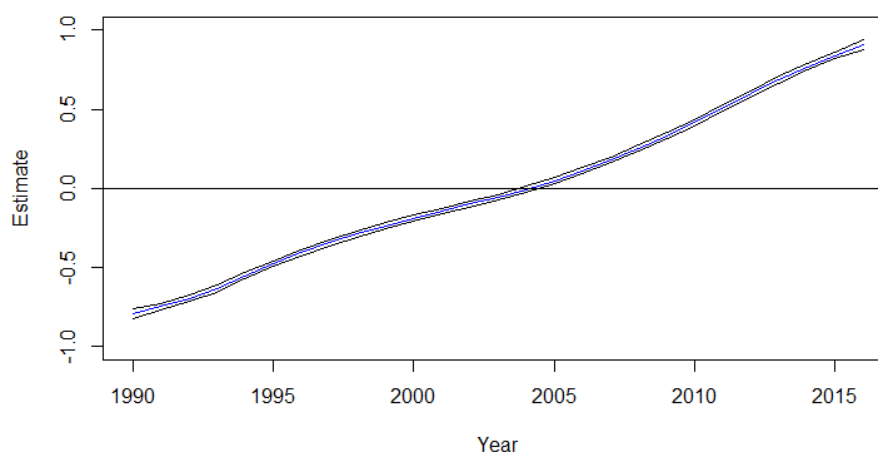


Figure 4.3: Temporal trend of the structured effect showing the posterior mean with 95% CI of the GHGs emissions from the industrial sector.

4.3.2 Spatio-temporal Plot. Figure 4.4 shows the effect of the interaction of space and time on the GHGs emissions from the industrial sector of each country over the period. Between 1990 to 2007, only eight countries were estimated to emit GHGs gases that belonged to the top category of emissions, with five out of the eight countries coming from Northern Africa. The countries are Morocco, Algeria, Tunisia, Libya, Egypt, Nigeria, Cameroon, and South Africa.

Interestingly, unlike the agricultural sector where it was observed that only Egypt from Northern Africa had a high estimate of GHGs emissions, all countries in the Northern Africa except for Mauritania had high estimates of GHGs emissions. Compared to other regions, Northern Africa and Southern Africa have the largest industrial base. The industrial sector mostly contributes to the gross domestic product of the countries in Northern Africa with industrial intensity highest in Tunisia, and Morocco, and Egypt having the largest industrial base in Northern Africa (de Alba, 2014).

Also, from Figure 4.4, it is seen that majority of the African countries have low estimated GHGs emissions throughout the study period. This finding may be attributed to the underdevelopment of the industrial sector of Africa stated in the 2013 annual report of the United Nations Industrial Development Organization (UNIDO).

A transition is seen in the industrial GHGs emissions of Senegal and Kenya from 2008 all through to 2016, as their estimated emissions transited to the top category. A similar case is observed for Ghana, Cote d'Ivoire, and Sudan in 2012 through 2016, Ethiopia and DRC in 2013 and 2015, respectively, through 2016.

Compared to other countries, most of the countries from Northern Africa and few countries from East and West Africa, and one country from Central and Southern Africa have high estimates of GHGs emissions from their industrial sector.

4.4 Energy Sector

4.4.1 Model-Based Temporal Trend. Figure 4.5 presents the structured temporal pattern of GHGs emissions from the energy sector. The behavior of the total GHGs emissions from the 44 African countries over the 27 years is revealed. Starting from 1990, a continuous increase is seen in the emissions throughout the time frame.

4.4.2 Spatio-temporal Plot. The effect of the interaction of space and time on the GHGs emissions and their resulting behavior from the energy sector of each country over the period is shown in Figure 4.6. It is observed that from 1990, the countries with an estimate of GHGs emissions in the top category from the energy sector are Morocco, Tunisia, Algeria, Libya, Egypt, Nigeria, and South-Africa through 2016.

From 2000-2001, the estimates of GHGs from Central Africa were among the top emissions but were followed by fluctuations between 2002-2006. The estimates maintained the category of high emissions from 2007 through 2016. Ethiopia, Angola, Tanzania, and Sudan are seen to have estimated GHGs among the top emissions in the year 2001, 2002, 2011, and 2013 respectively through 2016. A similar case is observed for Kenya and Ghana in the year 2013 through 2016.

Compared to other countries, most of the countries from Northern Africa, Nigeria and Ghana from West Africa, South Africa and Angola from Southern Africa, the Central African Republic from Central Africa, and four countries from East Africa have estimates of GHGs emissions to be among the top emissions from their energy sector. Interestingly, Mauritania, Mali, Niger, Chad, Congo, Burkinafaso, Togo, Benin, Namibia, and Somalia are countries whose energy sector least contributed to GHGs emissions in Africa



Figure 4.4: Spatio-temporal plot of the posterior mean of the GHGs emissions from the industry sector with spatially structured and temporally structured interaction.

during the early years later experienced a steady increase in their GHGs emissions over the 27 years.

The Generation of electricity is one of the major activities in the energy sector that produce GHGs emissions. There are various processes employed in generating electricity such as biomass, hydroelectric power, nuclear power, coal, gas and oil, wind, and solar. While hydroelectric power, wind, and solar produce little GHGs emissions, nuclear power produces no GHGs emissions. Biomass, coal, gas, and oil means of electricity generation produces a huge amount of GHGs emissions.

Oil and gas are mostly used in Northern and West Africa with Nigeria having the largest oil and gas plant. Coal is mostly employed in Southern Africa with South Africa having the largest coal plant, and hydroelectric are dominant in East and Central Africa (UNIDO, 2009). Furthermore, another contributor to GHGs emissions from this sector is the burning of wood and charcoal for household purpose especially cooking. There is the epileptic power supply in most African countries with some parts having no access to electricity especially the rural areas. People from such category rely mostly on firewood and charcoal

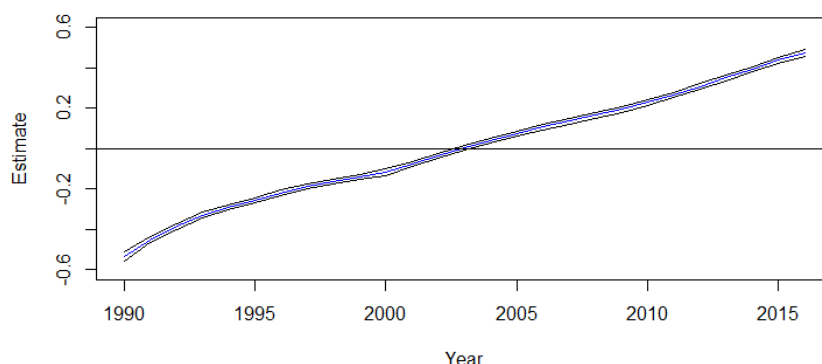


Figure 4.5: Temporal trend of the structured effect showing the posterior mean with 95% CI of the GHGs emissions from the energy sector.

for cooking (UNIDO, 2009).

Another huge contribution to GHGs emissions is transportation. Large population coupled with poor road network leads to serious traffic congestion and most vehicles across Africa that run on fuel especially petrol and diesel have low fuel efficiency (Hogarth et al., 2015). The combustion of these fuels releases GHGs into the atmosphere. Generally, GHGs emissions from the energy sector of the countries in Africa are increasing steadily over time with more countries having a high tendency to join the train of top emitters over time.

4.5 Land-Use Change and Forestry Sector

4.5.1 Model-Based Temporal Trend. Figure 4.7 presents the structured temporal pattern of GHGs emissions from the land-use sector of all the 44 African countries. A stable emission is seen between 1990 and 1996 and is followed by a continuous decrease in the emissions from 1997 to 2012. From the year 2012, a steady increase is observed up until 2016.

Unlike the other sectors which experienced a significant increase in their GHGs emission throughout the study period, land-use change and forestry sector experienced a decrease in its emissions. The result of the structured temporal trend can be explained by a report compiled by Hogarth et al. (2015). The report stated that between 2000 and 2010, there was a decrease in annual deforestation in sub-Saharan Africa as compared to the deforestation between 1990 and 2000.

One of the major activities emanating from the land-use change and forestry sector is deforestation. Forests help in removing CO_2 from the atmosphere as trees make use of CO_2 during photosynthesis and also, store up carbon. When the cut-down trees are set on fire, CO_2 is formed in this process as oxygen atoms mix up with a single carbon atom which in turn is released into the atmosphere.

4.5.2 Spatio-temporal Plot. Figure 4.8 shows the effect of the interaction of space and time on the GHGs emissions from the land-use change and forestry sector of each country over the study period. The behavior of the emissions across space and time are revealed. It is observed that from 1990-2016, countries that maintained the category of top emitters are Nigeria, Cameroun Angola, Sudan, DRC,

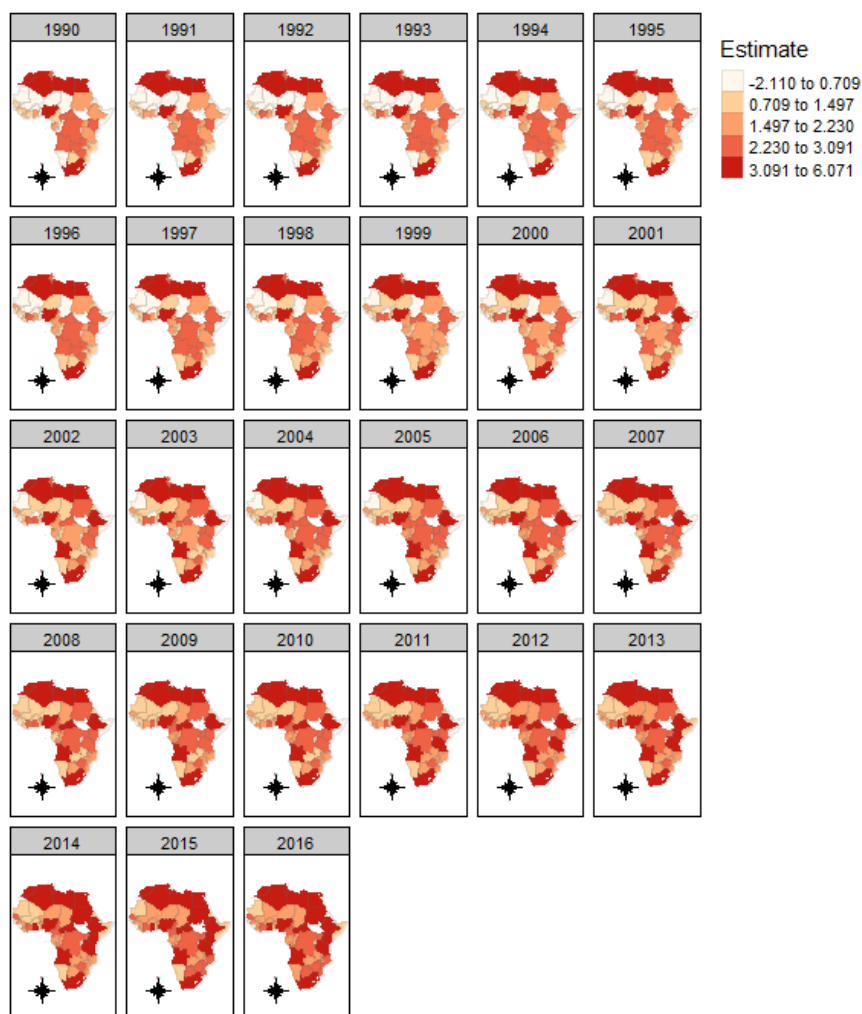


Figure 4.6: Spatio-temporal plot of the posterior mean of the GHGs emissions from the energy sector with spatially structured and temporally structured interaction.

Tanzania, and Zambia.

From 1990-1998, Kenya was among the top emitters of GHGs from their land-use change and forestry sector but dropped to the category of the least contributors from 1999-2016. Ghana joined the category of top emitters in the year 2002 and maintained this category up until 2016.

Fluctuations are observed in the emissions from the land-use change and forestry sector of Botswana, Mozambique, and Zimbabwe. Botswana attained and maintained the category of top emitters between 1990-2006, which was followed by a decrease from the year 2007-2013 and was restored to the top emitter category in 2014 up until 2016. Frequent fluctuations are noticed in Mozambique and are among the top emitters in the year 2016. Zimbabwe was among the top emitters from 1998-2000 but experienced a decrease in emissions from its land-use change and forestry sector until 2016.

Food and Agriculture Organization of the United Nations reported that the highest global deforestation in 2005 was from Nigeria. This was attributed to the clearing of forests for agricultural purposes and logging. Among all the countries in Africa, Nigeria is the most populated and implies that there is a

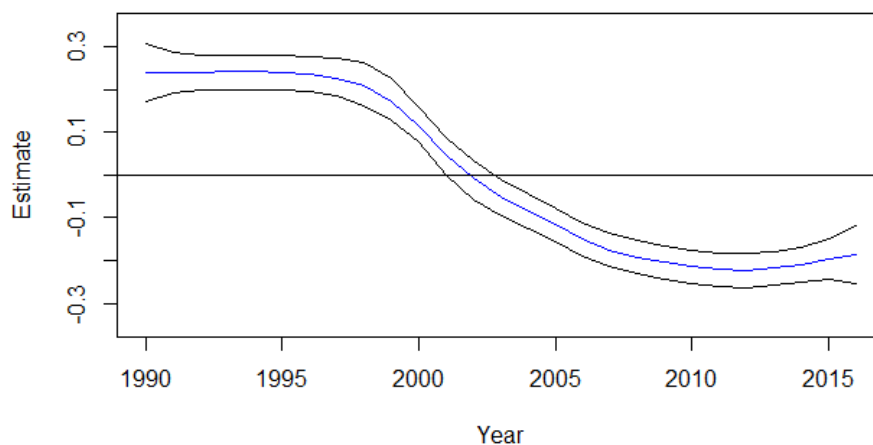


Figure 4.7: Temporal trend of the structured effect showing the posterior mean with 95% CI of the GHGs emissions from the land-use change and forestry sector.

very high demand for food to survive. Also, Botswana, Zimbabwe, and Tanzania recorded the highest amount of deforestation in 2011 among the Southern African countries.

Naidoo et al. (2013) reported that the countries in Southern Africa with the highest capacity of timber production are Zambia, Mozambique, and Angola. The forests serve as sources of raw material for these countries. Mayaux et al. (2013) revealed in the spatial plot of Africa's rainforests between 1990 and 2000 that there was the frequent occurrence of deforestation in DRC, Nigeria, and Cameroon.

Study findings also reveal that countries whose land-use change and forestry sector least contribute to GHGs emissions in Africa are South-Africa, Egypt, Mali, Mauritania, Kenya, Tunisia, Morocco, Senegal, Malawi, Gabon, Equatorial Guinea, Rwanda, Liberia, Burundi, Eritrea, Djibouti, and Libya. Interestingly, all the countries in Northern Africa belong to the least contributors. This may be due to the little amount of forest body found in Northern Africa countries. For South Africa, Naidoo et al. (2013) reported that between 2005 and 2010, no deforestation was recorded.

Generally, there seems to be no drastic increase in GHGs emissions from the land-use change and forestry sector of each country over the years. The countries contributing most to the emissions at the beginning of the period are still the ones contributing most to the emissions from land-use change and forestry sector.

4.6 Waste Sector

4.6.1 Model-Based Temporal Trend. Figure 4.9 presents the structured temporal pattern of GHGs emissions from the waste sector and reveals that over the period, there has been a continuous increase in the emissions of the GHGs from the waste sector.

4.6.2 Spatio-temporal Plot.

Figure 4.10 shows the effect of the interaction of space and time on the GHGs emissions from the waste sector of each country over the period. It is observed that from 1990, countries that are estimated to

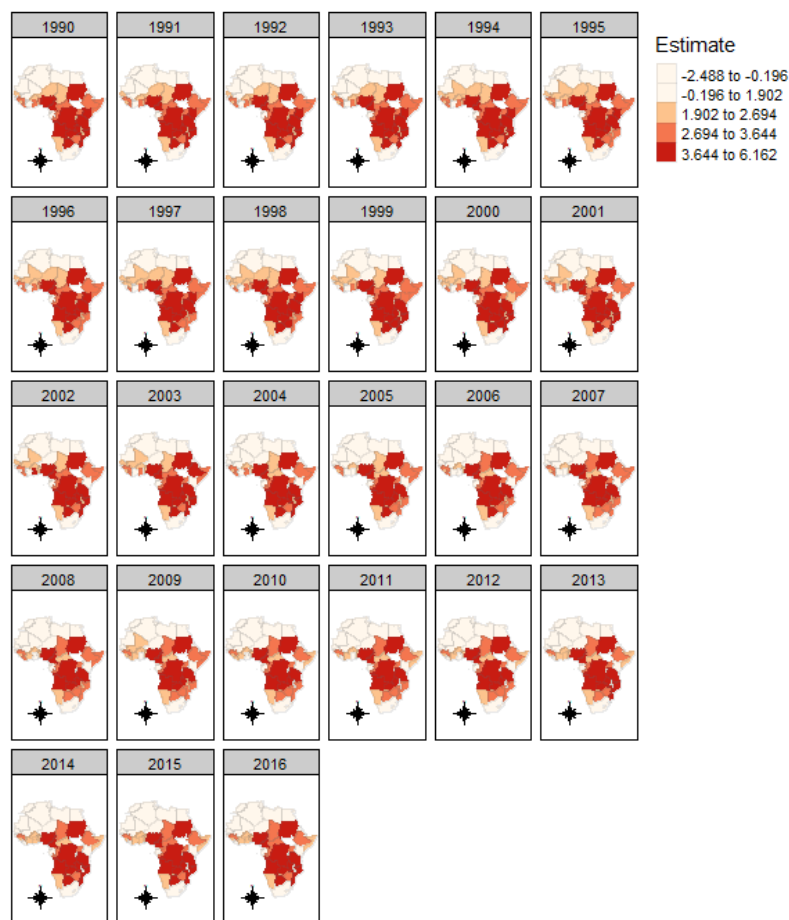


Figure 4.8: Spatio-temporal plot of the posterior mean of the GHGs emissions from the land-use change and forestry practices with spatially structured and temporally structured interaction of the 44 African countries.

emit a high amount of GHGs from their waste sector are Algeria, Egypt, Nigeria, DRC, and South-Africa through 2016. From 19902 to 2010, Ivory Coast was among the top emitters of GHGs from their waste sector but dropped out of the category since 2011.

The estimates of the GHGs of Tanzania, Morocco, Ethiopia, and Niger were among the top emissions in the years 1995, 1999, 2000, and 2005 respectively through 2016. A similar case is observed in Senegal, Cameroun, and Mozambique in 2013 up until 2016, Ghana, and Sudan in 2016.

Countries with high estimated GHGs emissions are mostly from sub-Saharan Africa with two countries from Northern Africa. [Godfrey et al. \(2019\)](#) reported that in 2012, 65% of solid waste generated in Africa emanated from sub-Saharan Africa and was attributed to poor waste management. For instance, there are enough open dumpsites, waste by the roadsides, and in the gutters in Nigeria and Ghana. Burning and decomposition of these wastes release GHGs into the atmosphere.

Countries whose waste sector least contributes to GHGs emissions in Africa are Libya, Mauritania, Mali, Uganda, Burundi, Rwanda, Kenya, Eritrea, Djibouti, Burkinafaso, Togo, Benin, Angola, Namibia, and Botswana. Generally, it is evident that over time, there has been an increase in the number of countries whose waste sector contributes more to GHGs emissions in Africa.

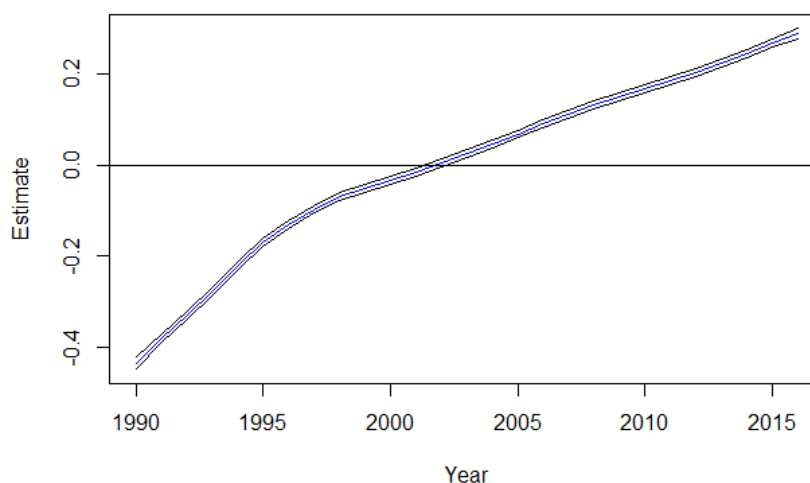


Figure 4.9: Temporal trend showing the posterior mean with 95% CI of GHGs emissions from the waste sector

4.7 Spatio-temporal Analysis of GHGs Emissions

4.7.1 Temporal Trend of GHGs Emissions. Figure 4.11 presents the temporal trend of GHGs emissions. It can be observed that all countries have positive trends, except that of three countries, Gabon, Kenya, and Rwanda, that experienced a fall in the GHGs emissions at some time. While an increasing trend of GHGs emissions over the years is observed for Algeria, Cameroun, Chad, Egypt, Ethiopia, Guinea, Mali, Morocco, Niger, Tunisia, and Tanzania, fluctuations in the GHGs emissions of the remaining countries are noticed over the years. Countries that constitute the top emitters of GHGs throughout 27 years are South Africa, Nigeria, Egypt, Zambia, Democratic Republic of Congo (DRC), and Cameroun.

4.7.2 Model-Based Temporal Trend. Figure 4.12 presents the structured temporal pattern for GHGs emissions. The emissions over the time frame have a pattern similar to a sinusoidal pattern. An increase in GHGs emissions is visible between 1990 and 1998 followed by a decline from 1999 to 2003. GHGs emissions picked up again in 2003 with a continuous increase until 2008. Little or no change in the total GHGs emissions between the years 2008 and 2010 is seen, followed by a slight reduction in GHGs emissions from 2010 to 2012. Since then, an increase in GHGs emission is evident up to the year 2016.

4.7.3 Spatio-temporal Plot. Figure 4.13 shows the effect of the interaction of space and time on the total GHGs emissions from each country over time. It is observed that from 1990-2016, the estimates of the total GHGs emissions of Nigeria, DRC, Angola, Zambia, Tanzania, and South-Africa were among the top emissions.

The GHGs emissions estimate of Cameroun, Egypt, Sudan, Algeria, and Ethiopia, are among the top emissions in the years 1993, 1996, 2002, 2007, and 2008 respectively. The countries maintained this category up until 2016, except for Sudan, which fell out of the category between 2013 to 2015 and later



Figure 4.10: Spatio-temporal plot of the posterior mean of the GHGs emissions from the waste sector with spatially structured and temporally structured interaction of the 44 African countries.

bounced back in 2016.

Countries that least contributes to GHGs emissions in Africa are Mauritania, Gabon, Equatorial Guinea, Rwanda, Eritrea, Guinea-Bissau, Sierra-Leone, Burundi, and Djibouti. Interestingly, it is noticed that there was a drastic reduction in GHGs emissions from Kenya between the years 2000-2006, and there has been an increase afterward.

Generally, the total GHGs emissions in Africa are increasing steadily within each country, with more countries having a high tendency to join the train of top emitters over time.

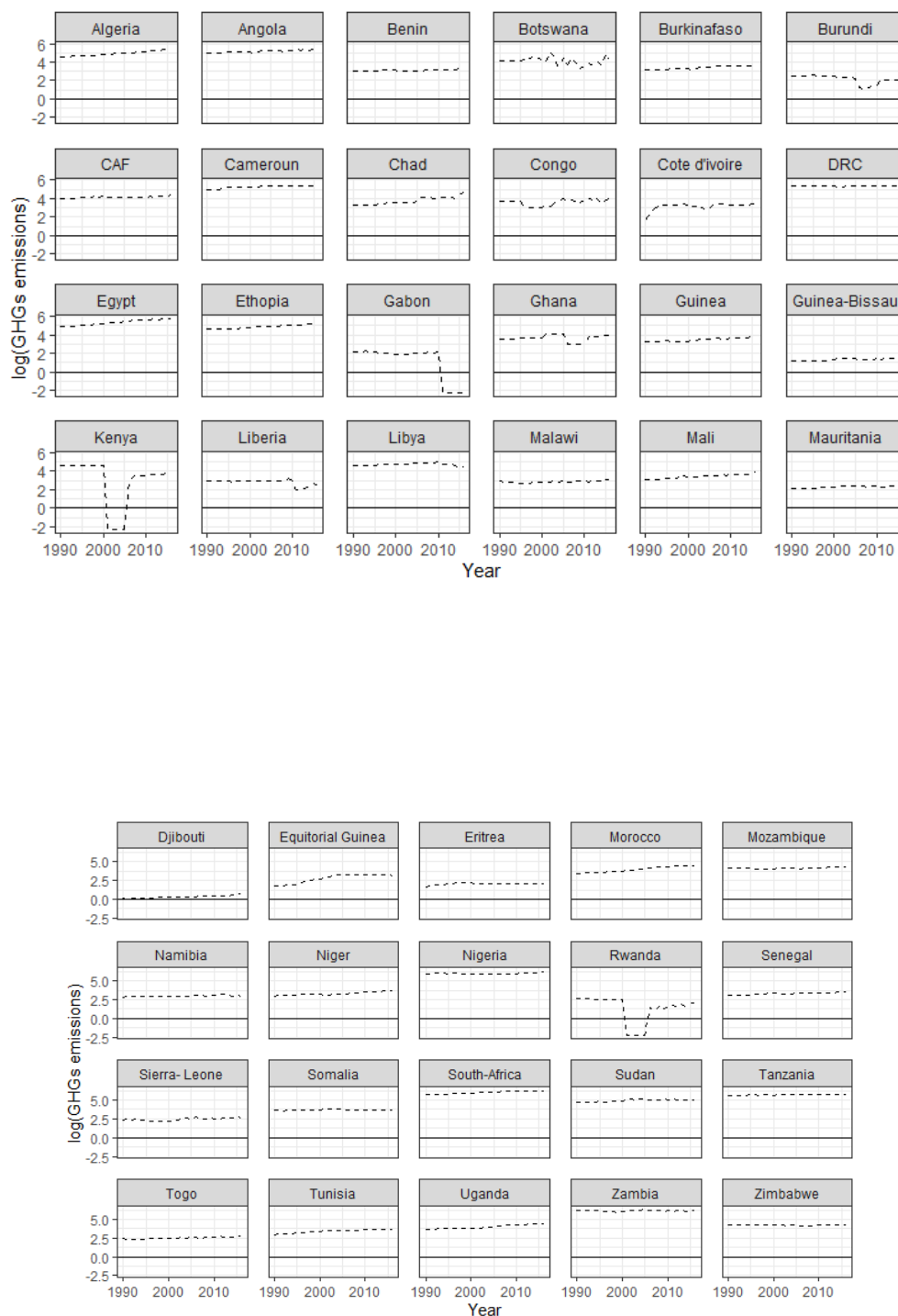


Figure 4.11: Temporal trend plot of the log of the GHGs emissions for the 44 African countries.

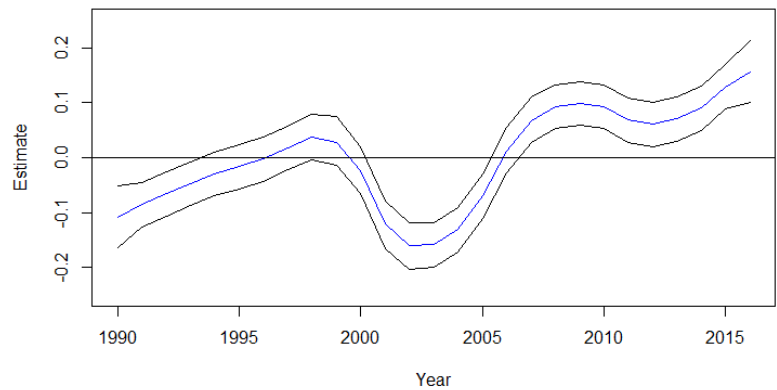


Figure 4.12: Structured Temporal trend showing the posterior mean with 95% CI of the GHGs emissions

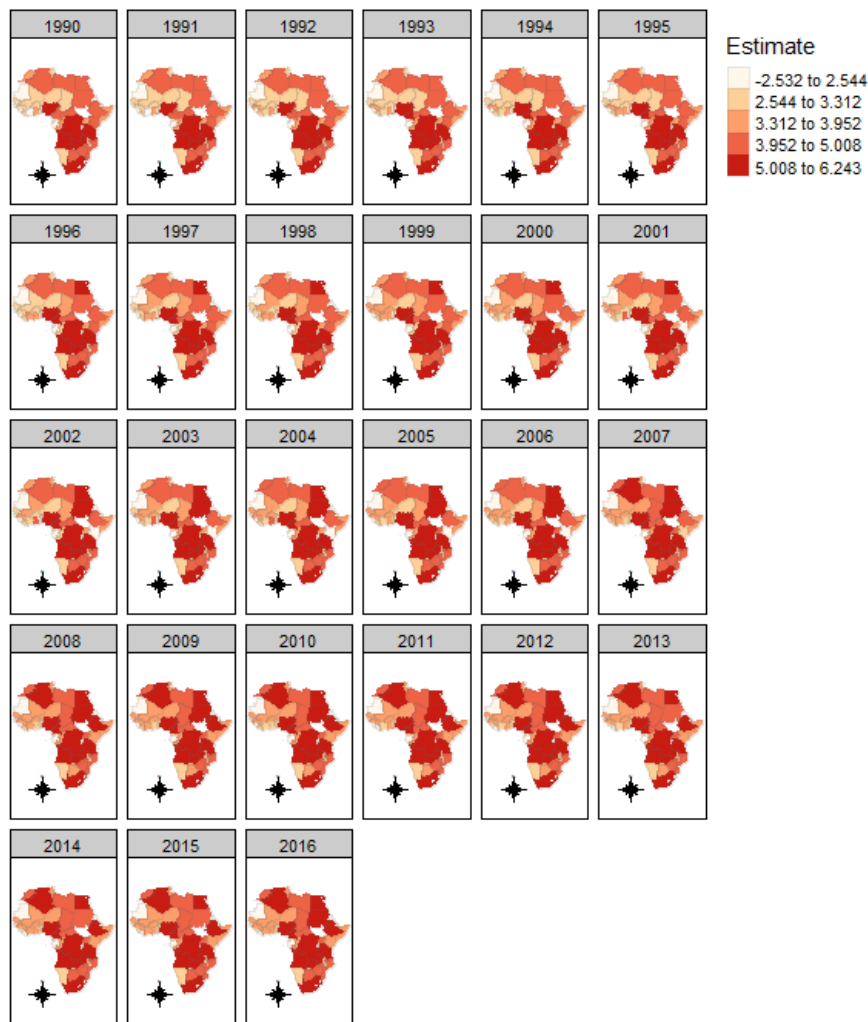


Figure 4.13: Spatio-temporal plot of the posterior mean of the GHGs emissions with spatially structured and temporally structured interaction.

5. Conclusion

The importance of studying and understanding the behavior of GHGs emissions cannot be overlooked as they contribute to global warming and global warming in turn, contributes to climate change. This study analyzed the patterns in the emissions of GHGs employing Bayesian Spatio-temporal approach that took into consideration, not just space and time effect, but also the interaction between these two entities.

The different forms that the spatial, temporal, and interactions effect could take were considered in the models that were developed for the study. Interestingly, the model that included the spatial structure and temporal structure interaction performed better than the other models.

The study findings revealed that spatial correlation and temporal correlation exist in the emissions of GHGs from Africa across the 27 years. The temporal correlation that exists may be as a result of the life-span of these GHGs in the atmosphere. According to IPCC, the lifetime of CO_2 in the atmosphere is between 5 to 200 years while the lifetime of CH_4 , N_2O , SF_6 , PFCs, and HFCs in the atmosphere are 12, 114, 3200, 50000, and 270 years, respectively.

Spatio-temporal plots were generated as a result of the space-time interaction considered in the model. The plots revealed that the effect of the boundaries shared among countries on the GHGs emissions became more evident over a long period compared to a short period.

Furthermore, the study findings showed that the countries in Africa with total GHGs emissions that belonged to the category of top emissions are Nigeria, DRC, Angola, Zambia, Tanzania, South-Africa, Egypt, Ethiopia, Cameroun, Sudan, and Algeria. The study considered the sectors that contribute to these emissions in Africa. It is discovered that the estimated GHGs emissions from the agricultural sector of Nigeria, DRC, Angola, Zambia, Tanzania, South-Africa, Egypt, Ethiopia, and Sudan are top.

For the industrial sector, Nigeria, DRC, Angola, South-Africa, Egypt, Ethiopia, Cameroun, and Sudan have top estimated GHGs emissions. Nigeria, Angola, Tanzania, South-Africa, Egypt, Ethiopia, Sudan, and Algeria have top GHGs emissions estimate from their energy sectors. The countries with a top estimate of GHGs emissions from the land-use change and forestry sectors are Nigeria, DRC, Angola, Zambia, Tanzania, Ethiopia, Cameroun, and Sudan. In contrast, the countries from the waste sectors are Nigeria, DRC, Tanzania, South-Africa, Egypt, Ethiopia, Cameroun, Sudan, and Algeria.

Interestingly, the two countries with estimated GHGs emissions that belonged to the top emissions from all the five sectors stated are Nigeria from West Africa and Sudan from East Africa. Future work may be done to study the individual sector of the countries that are generally top emitters of GHGs and possibly include factors that may likely influence GHGs emissions such as population size, land size, gross domestic product, region, and so on as covariates in the Bayesian Spatio-temporal model.

Appendix

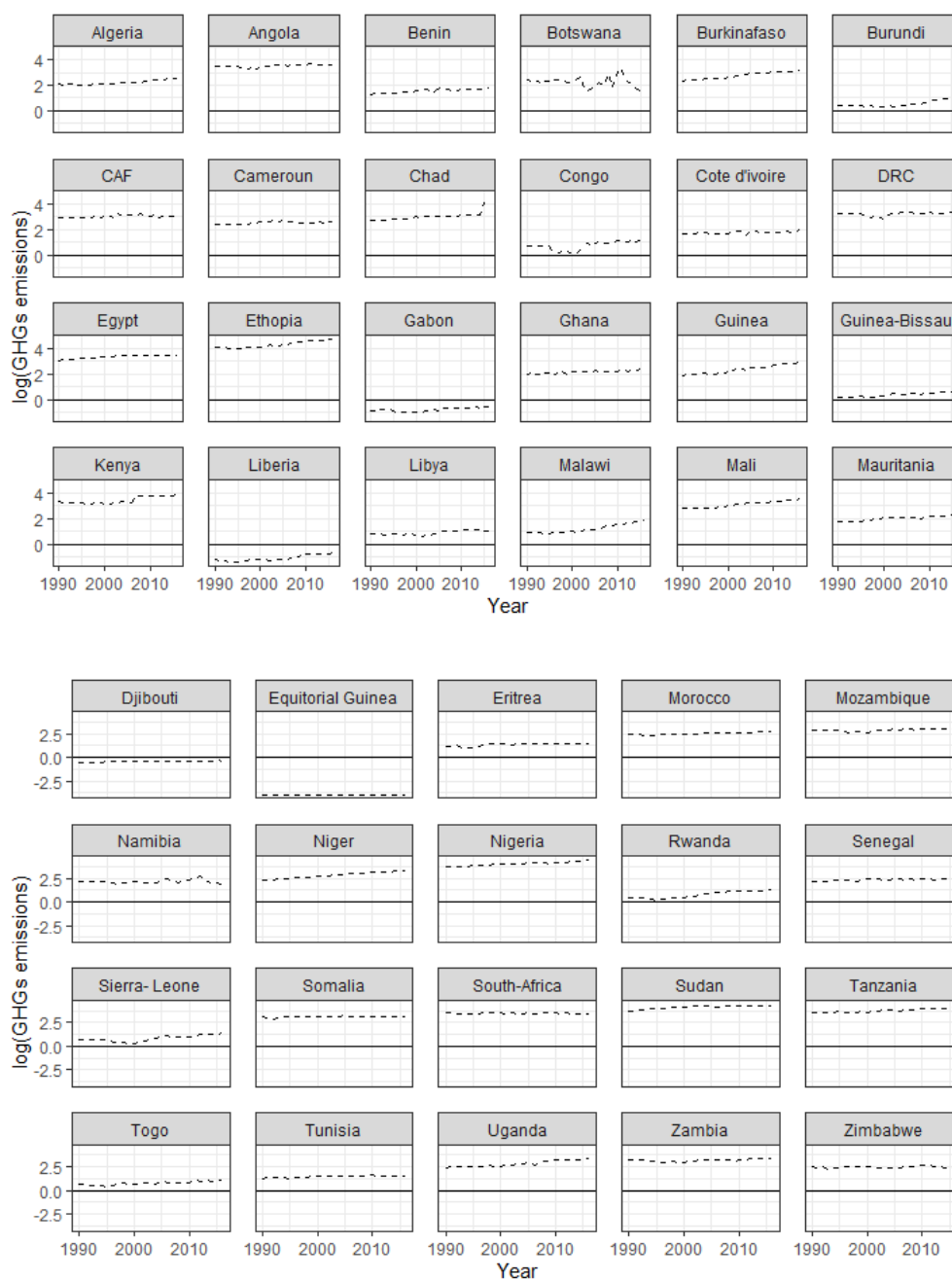


Figure 5.1: Temporal trend plot of the log of GHGs emissions from the agriculture sector



Figure 5.2: Temporal trend plot of the log of the GHGs emissions from the industry sector of the 44 African countries.



Figure 5.3: Temporal trend plot of the log of the GHGs emissions from the energy sector for the 44 African countries.

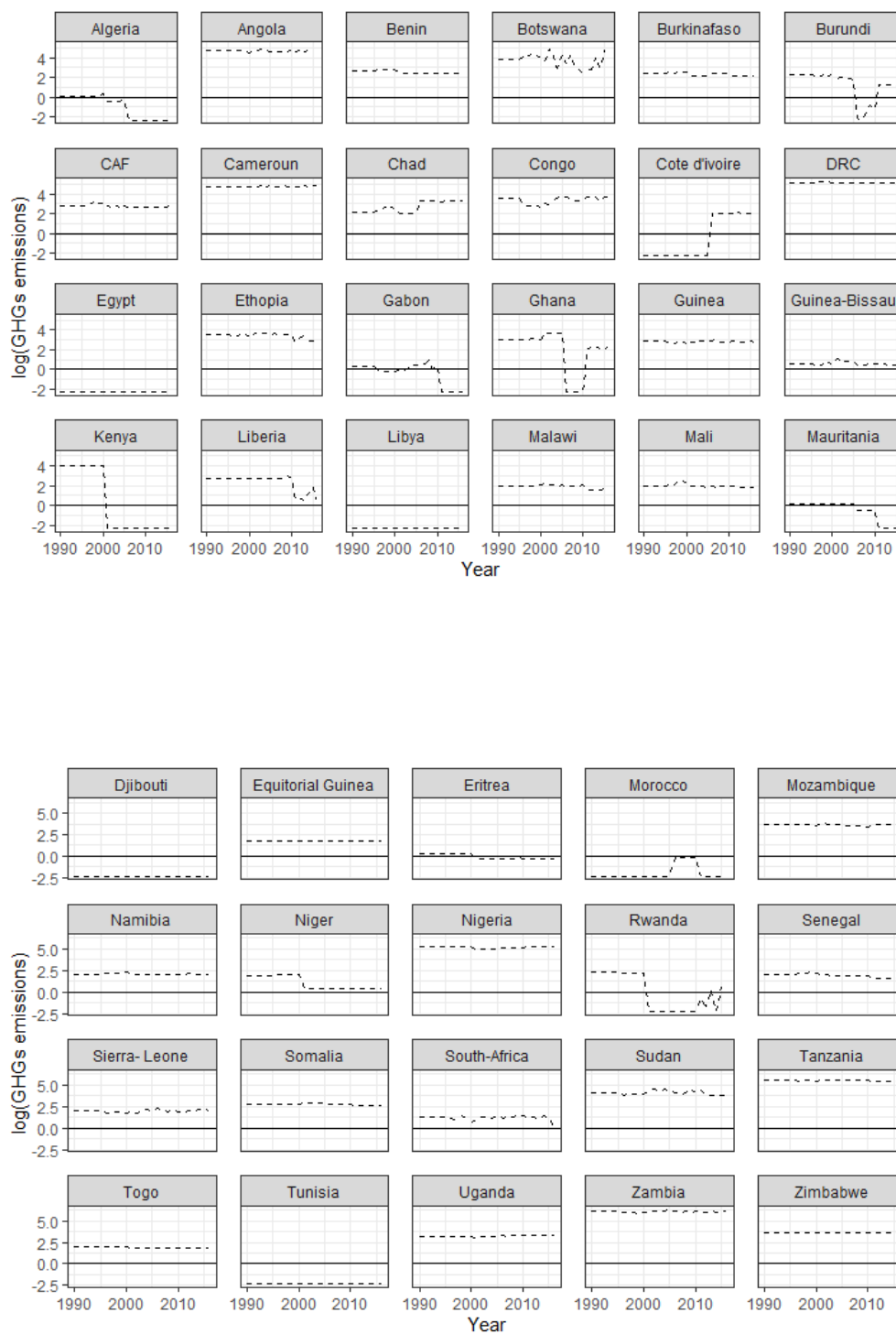


Figure 5.4: Temporal trend plot of the log of the GHGs emissions from land-use change and forestry practices of the 44 African countries.



Figure 5.5: Temporal trend plot of the log of GHGs emissions from the waste sector

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